Computer Graphics Quiz Practice

DDA Line Algorithm & Bresenham's Midpoint Line Algorithm

Fall 2026–2027 • Student Practice + Instructor Answer Key

Course	Computer Graphics	**Practice Type**	Written Quiz Preparation
Topics	DDA + Midpoint Line		

Scope and instructions

• This practice set is based on the supplied Scan Conversion lecture material only.

• No Midpoint Circle Algorithm is included.

• Students should be able to write the DDA and Midpoint Line algorithms, show intermediate calculations, and identify the generated pixels.

• For Midpoint Line questions, follow the lecture convention: for $m < 1$, $d > 0$ selects NE and $d \leq 0$ selects E; for $m > 1$, $d > 0$ selects NE and $d \leq 0$ selects N.

Part A — Quick Study Reminder

Topic	Key formula / rule	What students should remember
DDA	$m = dy/dx$	Increment one driving coordinate and add the appropriate fractional increment to the other coordinate.
DDA, $m < 1$	$x(i+1)=x(i)+1$; $y(i+1)=y(i)+m$	Round only for the displayed pixel.
DDA, $m > 1$	$x(i+1)=x(i)+1/m$; $y(i+1)=y(i)+1$	Y is the driving coordinate.
Midpoint, $m < 1$	$dstart = 2dy - dx$	$d > 0 \rightarrow$ NE; $d \leq 0 \rightarrow$ E.
Midpoint, $m > 1$	$dstart = 2dx - dy$	$d > 0 \rightarrow$ NE; $d \leq 0 \rightarrow$ N.

Part B — Practice Questions

Q1. Foundations of scan conversion **[4 marks]**

a) Define rasterization / scan conversion in the context of line drawing.

b) What is an incremental line algorithm?

c) State the three requirements listed in the lecture for an incremental algorithm.

Q2. Write the DDA algorithm for a shallow line **[6 marks]**

A line has $0 < m < 1$. Write the DDA line algorithm in mathematical steps. Your answer must show how to calculate m and how to obtain the next pixel from the current one. No programming code is required.

Q3. DDA numerical problem — lecture-style example [8 marks]

Use the DDA Line Algorithm to rasterize the line from P0 = (4,5) to Pn = (7,7).

Show: dx, dy, slope m, the unrounded coordinate at every step, the rounded pixel selected at every step, and the final pixel sequence.

Q4. DDA numerical problem — steep slope [8 marks]

Use the lecture DDA rule for m > 1 to rasterize the line from (2,2) to (4,7).

Show the slope, identify the driving coordinate, calculate 1/m, and list each selected pixel. Round the non-driving coordinate to the nearest pixel.

Q5. DDA limitations [4 marks]

Write any two disadvantages of the DDA Line Algorithm discussed in the lecture and briefly explain each one.

Q6. Write the Midpoint Line algorithm for m < 1 [8 marks]

For 0 < m < 1, write the complete Bresenham / Midpoint Line Algorithm in mathematical form.

Your answer must include dx, dy, dstart, the E and NE decisions, coordinate updates, and decision-parameter updates.

Q7. Midpoint derivation — where does dstart come from? [8 marks]

Starting from y = mx + B and F(x,y) = ax + by + c, show how the lecture obtains dstart = 2dy − dx for 0 < m < 1.

Your derivation should identify a, b and c, evaluate the first midpoint, and explain why the decision expression is multiplied by 2.

Q8. Midpoint numerical problem — lecture example [10 marks]

Apply the Midpoint Line Algorithm to draw the line from (4,5) to (7,7).

Show dx, dy, dstart, the two possible decision increments, every E/NE choice, every updated d value, and the final pixel sequence.

Q9. Write the Midpoint Line algorithm for m > 1 [8 marks]

For m > 1, write the complete mathematical algorithm using N and NE as the candidate pixels.

Include dstart, the N/NE decision rule, coordinate updates and decision-parameter updates.

Q10. Midpoint numerical problem — steep line [10 marks]

Use the m > 1 Midpoint Line Algorithm to rasterize the line from (2,2) to (4,7).

Show every calculation and prepare a table containing the current decision value, selected direction, and new pixel.

Q11. Compare the two line algorithms **[6 marks]**

Compare DDA and Bresenham's Midpoint Line Algorithm under the following points: arithmetic used, rounding, incremental nature, possible numerical error, and pixel-selection idea.

Q12. Short reasoning questions **[10 marks]**

Answer each in one or two sentences:

a) Why is x the driving coordinate when $0 < m < 1$?

b) Why is y the driving coordinate when $m > 1$?

c) In the lecture convention for $m < 1$, what does $d > 0$ mean for the next pixel?

d) Why is multiplying the decision parameter by 2 allowed?

e) Why does the Midpoint algorithm not need a round() operation for its core pixel decision?

Instructor Answer Key / Self-Study Solutions

Use after attempting the questions first.

A1. Foundations of scan conversion

Rasterization / scan conversion: The process of converting geometric/vector descriptions into discrete raster pixels for display or storage.

Incremental algorithm: A line-drawing method that begins at an initial pixel and determines each next pixel using calculations based on the preceding step.

Lecture requirements: Integer pixel grid, slope, and disjoint pixel selection.

A2. DDA algorithm for 0 < m < 1

Given P0=(x0,y0) and Pn=(xn,yn):

1) dx = xn − x0 and dy = yn − y0.

2) m = dy/dx.

3) Start with xi=x0 and yi=y0.

4) For 0 < m < 1, increment x by 1: x(i+1)=xi+1.

5) Update y using y(i+1)=yi+m.

6) Display the pixel (x(i+1), round(y(i+1))).

7) Continue until the destination x-coordinate is reached.

Important: keep the unrounded y value for the next calculation; round only the pixel that is displayed.

A3. DDA numerical problem: (4,5) to (7,7)

dx=7−4=3; dy=7−5=2; m=dy/dx=2/3≈0.67. Since m<1, x increases by 1 and y increases by m each step.

Step	x	unrounded y	selected pixel
Start	4	5.000	(4,5)
1	5	5.667	(5,6)
2	6	6.334	(6,6)
3	7	7.001 (≈7)	(7,7)

A4. DDA numerical problem: (2,2) to (4,7)

dx=2, dy=5, so m=5/2=2.5>1. Therefore y is the driving coordinate.

1/m=0.4. Use x(i+1)=xi+0.4 and y(i+1)=yi+1, rounding x only for the displayed pixel.

Step	unrounded x	y	selected pixel
Start	2.0	2	(2,2)
1	2.4	3	(2,3)
2	2.8	4	(3,4)
3	3.2	5	(3,5)
4	3.6	6	(4,6)
5	4.0	7	(4,7)

A5. DDA limitations

Any two source-supported points:

• Floating-point arithmetic is time-consuming relative to integer-only updates.

• Successive floating-point additions can accumulate round-off error and make a long rasterized line drift from the ideal path.

• The lecture also notes orientation dependence and poorer endpoint accuracy.

A6. Midpoint Line algorithm for 0 < m < 1

Calculate dx=xn−x0 and dy=yn−y0.

Initial decision parameter: dstart=2dy−dx.

If d>0: select NE; x(i+1)=xi+1, y(i+1)=yi+1, and dnew=dold+2(dy−dx).

If d≤0: select E; x(i+1)=xi+1, y(i+1)=yi, and dnew=dold+2dy.

Repeat until the final x-coordinate is reached.

A7. Derivation of dstart=2dy−dx

Start with y=mx+B and m=dy/dx, so y=(dy/dx)x+B.

Multiply by dx and rearrange: dy·x − dx·y + Bdx = 0.

Therefore F(x,y)=ax+by+c with a=dy, b=−dx and c=Bdx.

For current pixel (x0,y0), the first midpoint between E and NE is M=(x0+1, y0+1/2).

dstart = F(M) = a(x0+1)+b(y0+1/2)+c.

Because the initial pixel lies on the ideal line, F(x0,y0)=0. Hence dstart = a+b/2 = dy−dx/2.

Multiplying by positive 2 preserves the sign used for the decision and removes the fraction, giving the scaled parameter dstart=2dy−dx.

A8. Midpoint numerical problem: (4,5) to (7,7)

dx=3; dy=2; m=2/3<1.

dstart=2dy−dx=4−3=1.

If E is chosen, add 2dy=4. If NE is chosen, add 2(dy−dx)=2(2−3)=−2.

Step	current d	decision	new pixel	new d
Start	1	—	(4,5)	1
1	1	NE	(5,6)	1−2 = −1
2	−1	E	(6,6)	−1+4 = 3
3	3	NE	(7,7)	End

A9. Midpoint Line algorithm for m > 1

Calculate dx=xn−x0 and dy=yn−y0.

Initial decision parameter: dstart=2dx−dy.

If d>0: select NE; x(i+1)=xi+1, y(i+1)=yi+1, and dnew=dold+2(dx−dy).

If d≤0: select N; x(i+1)=xi, y(i+1)=yi+1, and dnew=dold+2dx.

Repeat until the final y-coordinate is reached.

A10. Midpoint numerical problem: (2,2) to (4,7)

dx=2; dy=5; m=2.5>1.

dstart=2dx−dy=4−5=−1.

N update: +2dx=+4. NE update: +2(dx−dy)=2(2−5)=−6.

Step	current d	decision	new pixel	new d
Start	−1	—	(2,2)	−1
1	−1	N	(2,3)	−1+4 = 3
2	3	NE	(3,4)	3−6 = −3
3	−3	N	(3,5)	−3+4 = 1
4	1	NE	(4,6)	1−6 = −5
5	−5	N	(4,7)	End

A11. DDA vs Midpoint Line Algorithm

Point	DDA	Midpoint / Bresenham
Arithmetic	Fractional / floating-point increments	Primarily integer arithmetic after setup
Rounding	Required for raster pixel output	Core decision does not use round()
Incremental?	Yes	Yes
Numerical issue	Successive floating-point error may accumulate	Tracks an integer decision/error parameter
Pixel selection idea	Generate continuous next coordinate, then round	Choose between two candidate pixels using a midpoint decision

A12. Short reasoning answers

a) For 0<m<1, the horizontal change dominates; advancing x one unit at a time avoids skipping columns.

b) For m>1, the vertical change dominates; advancing y one unit at a time avoids skipping rows.

c) Under the lecture convention for m<1, d>0 selects the NE pixel.

d) The algorithm only uses the sign of d. Multiplying by positive 2 does not change that sign and removes the 1/2 fraction.

e) The Midpoint algorithm directly chooses an integer candidate pixel using the decision parameter rather than generating a fractional coordinate and rounding it.

One-Page Formula Sheet for Revision

Case	Initial / update formulas	Decision
DDA, m<1	xnext=x+1; ynext=y+m	Round y only for plotted pixel
DDA, m>1	xnext=x+1/m; ynext=y+1	Round x only for plotted pixel
Midpoint, m<1	d0=2dy−dx; E: d+=2dy; NE: d+=2(dy−dx)	d>0 → NE; d≤0 → E
Midpoint, m>1	d0=2dx−dy; N: d+=2dx; NE: d+=2(dx−dy)	d>0 → NE; d≤0 → N